

Write legibly showing all the steps VERY clearly. Max. score=100 (Total points = 105).

- (1) (a) Let $(\Omega, \mathcal{F}, P) = ([0, 1], \mathcal{B}([0, 1]), m)$, where m denotes the Lebesgue measure on $[0, 1]$. Define the sequence of random variables X_n such that $X_1 \equiv 0$, $X_2 \equiv -5$ and $X_n(\omega) = n^2 \mathbf{I}_{[1-\frac{1}{n^2}, 1]}(\omega)$, $\omega \in \Omega$, $n \geq 3$. Let $X \equiv 0$.

- (i) Prove that $X_n \rightarrow X$ a.e. (P) as $n \rightarrow \infty$.
- (ii) Prove: $\{X_n\}$ **not** uniformly integrable.
- (iii) Prove: X_n does **not** converge to X in L^1 as $n \rightarrow \infty$.

- (b) In general, under what conditions on the functions $\{X_n\}$ would

" $X_n \rightarrow X$ a.e (P)" imply " $X_n \rightarrow X$ in L^p " for $p \geq 1$?

((3×10)+10=40 points)

- (2) Let $(\Omega, \mathcal{F})$ be a measurable space, and μ_1, μ_2, ν are σ -finite measures on it. Define the set function: $\mu_3 \equiv 3\mu_1 + 2\mu_2$ (i.e., for all $A \in \mathcal{F}$, $\mu_3(A) = 3\mu_1(A) + 2\mu_2(A)$). Then, prove the following:

- (a) μ_3 is a measure.

- (b) If $\mu_1 \ll \nu$ and $\mu_2 \ll \nu$, then $\mu_3 \ll \nu$. Also, express the Radon-Nikodym derivative $\frac{d\mu_3}{d\nu}$ in terms of $\frac{d\mu_1}{d\nu}$ and $\frac{d\mu_2}{d\nu}$ (proof required).

Let $(\Omega, \mathcal{F}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ and $\mu = (0.3)\mu_1 + (0.7)\mu_2$, where $\mu_1 = \text{Normal}(2, 3)$ and $\mu_2 = \text{Poisson}(6)$. Let $\nu = m =$ the Lebesgue measure.

- (c) Find the Lebesgue decomposition of μ with respect to ν , and find the Radon Nikodym derivative $\frac{d\mu_a}{d\nu}$, where μ_a is the absolutely continuous component of μ in the decomposition.

- (d) Let X be a random variable with distribution μ (i.e. $P_X = \mu$). Find $E(X^2)$.
(Hint: a $\text{Normal}(2, 3)$ random variable has mean is 2, variance is 3 and a $\text{Poisson}(6)$ random variable has mean 6 and variance 6).

(10+(10+10)+10+10=50 points)

- (3) Suppose X is a random variable defined on some probability space $(\Omega, \mathcal{F}, P)$. Then show: $X \in L^{p_2}(\Omega, \mathcal{F}, P)$ implies $X \in L^{p_1}(\Omega, \mathcal{F}, P)$, where $0 < p_1 < p_2 < \infty$.
(Recall: $X \in L^p(\Omega, \mathcal{F}, P)$ means $E[|X|^p] < \infty$)

(15 points)

TABLE 4.6.1. Some discrete univariate distributions.

Mean μ	$\mu(A), A \in \mathcal{B}(\mathbb{R})$
Bernoulli (p), $0 < p < 1$	$\sum_{i=0}^1 p^i (1-p)^{1-i} I_A(i)$
Binomial (n, p), $0 < p < 1, n \in \mathbb{N}$	$\sum_{i=0}^n \binom{n}{i} p^i (1-p)^{n-i} I_A(i)$
Geometric (p), $0 < p < 1$	$\sum_{i=0}^{\infty} p(1-p)^i I_A(i)$
Poisson (λ), $0 < \lambda < \infty$	$\sum_{i=0}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!} \cdot I_A(i)$

TABLE 4.6.2. Some standard absolutely continuous distributions. Here m denotes the Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Measure μ	$\mu(A), A \in \mathcal{B}(\mathbb{R})$
Uniform (a, b), $-\infty < a < b < \infty$	$m(A \cap [a, b]) / (b - a)$
Exponential (β), $\beta \in (0, \infty)$	$\int_A \frac{1}{\beta} \exp(-x/\beta) I_{(0, \infty)}(x) m(dx)$
Gamma (α, β), $\alpha, \beta \in (0, \infty)$	$\int_A \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} \exp(-x/\beta) I_{(0, \infty)}(x) m(dx)$ where $\Gamma(a) = \int_0^\infty x^{a-1} e^{-x} dx, a \in (0, \infty)$
Beta (α, β), $\alpha, \beta \in (0, \infty)$	$\frac{1}{B(\alpha, \beta)} \int_A x^{\alpha-1} (1-x)^{\beta-1} I_{(0, 1)}(x) m(dx),$ where $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a+b), a, b \in (0, \infty)$
Cauchy (γ, σ) $\gamma \in \mathbb{R}, \sigma \in (0, \infty)$	$\int_A \frac{1}{\pi\sigma} \frac{\sigma^2}{\sigma^2 + (x-\gamma)^2} m(dx)$
Normal (γ, σ^2), $\gamma \in \mathbb{R}, \sigma \in (0, \infty)$	$\int_A \frac{1}{\sqrt{2\pi}\sigma} \exp(-(x-\gamma)^2 / 2\sigma^2) m(dx)$
Lognormal (γ, σ^2), $\gamma \in \mathbb{R}, \sigma \in (0, \infty)$	$\int_A \frac{1}{\sqrt{2\pi}\sigma} \frac{e^{-(\log x - \gamma)^2 / 2\sigma^2}}{x} I_{(0, \infty)}(x) m(dx)$